KAICHENG MA
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EDUCATION

Duke Kunshan University **Kunshan, China**
B.S. in Computation and Design - Computer Science, GPA: 3.719/4.0 *Expected May 2027*

- Honors: Dean's List (Spring 2024, Fall 2024, Fall 2025)
- Relevant Coursework: Introduction to Programming and Data Structures, Computer Organization and Programming, Discrete Mathematics for Computer Science, Probability and Statistics, Computation and Problem Solving, Introductory Psychology, Web-Based Multimedia Design, Interaction and Physical Computing, Introduction to Operating Systems, Integrated Science - Physics.

Duke University **Durham, NC**
Exchange Program *Jan 2026 – May 2026*

- Relevant Coursework: Design and Analysis of Algorithms, Linear Algebra

RESEARCH EXPERIENCE

Social-Cue-Elicited AI-VR Integrated Anxiety Relief System *Oct 2025 – Present*
Undergraduate Research Lead | Advisor: Prof. Yucheng Jin

- Designed a controlled VR study to comparing asynchronous human traces vs. synchronous AI companionship on stress recovery, examining the effects of different social cues and individual differences in user responses.
- Built an immersive 3D forest and interactive VR tasks in Unity; developed two experimental conditions with consistent environments, interactions, and audio to isolate social-cue effects.
- Developed an embodied AI companion using Unity, Python, and a Qwen LLM, incorporating user context, rule-based constraints, behavioral memory, and asynchronous decision-making to enable context-aware interactions.
- Conducted a counterbalanced within-subjects experiment with cognitive stress induction and VR recovery sessions; collected VAS-Stress, social connectedness, extraversion, and PPG-derived physiological measures to evaluate stress recovery and individual differences.
- Found significant subjective stress reductions in both conditions; after controlling for pre-VR stress, synchronous AI companionship showed no conclusive overall advantage, while higher extraversion was associated with greater relative benefits from synchronous companionship.

Multi-Agent Adaptive Game System *Aug 2024 – Aug 2025*
Undergraduate Research Team Member | Advisor: Prof. Bing Luo

- Developed a multi-agent adaptive game system where each agent acts as an NPC in an AI-driven autonomous world, enabling characters to interact, influence one another, and create more realistic gameplay.
- Designed an Intention AI self-decision mechanism that analyzes story progress, in-game time, location settings, and other contextual information to weigh options and generate optimal NPC decisions.
- Implemented an OCEAN-based personality-driven dialogue mechanism; used RoBERTa and pretrained models to improve personality expression accuracy, behavioral stability, and emotional consistency.
- Participated in building multi-NPC dialogue workflows with LangChain and LangGraph; iterated a Chain-of-Thought multi-stage intention mechanism (environment analysis, emotion update, intention generation, and filtering) and curated/labeled datasets for a personality analysis system.

PRODUCT & DESIGN EXPERIENCE

AI Voice Interaction and Haptic Feedback Companion Teddy Bear Prototype *Jan 2025 – Apr 2025*
Undergraduate Research Lead | Advisor: Prof. Giovanni Santini

- Led a three-person team in designing an intelligent companion teddy bear for student loneliness, stress, and sleep issues; conducted user surveys and literature/product reviews to inform design.
- Developed ESP32-based hardware in C++ for animated LCD expressions, warm lighting, and breathing-guided vibration; tuned haptic rhythms through iterative user testing and feedback.

- Coordinated voice-software and hardware integration and co-designed the enclosure; optimized display layout, managed 3D printing, and assembled the functional prototype.
- Independently redesigned the voice architecture, replacing fixed recordings and file-based exchanges with continuous INMP441 I²S audio capture on ESP32-S3 and USB PCM streaming using COBS framing, sequence numbers, and CRC32 checks.
- Integrated PocketSphinx wake-word detection, VAD, Qwen speech recognition and dialogue, and TTS with bidirectional JSON hardware control, enabling continuous multi-turn conversations, voice interruptions, and AI-driven physical responses.

Social Matching Device and App for College Student Friendship *Mar 2025 – Jun 2025*

Project Lead

- Led primary research (international student interviews, surveys) and secondary research (social media, news, academic articles) to identify friendship pain points and design requirements among international college students.
- Designed device and app concepts using Procreate, Figma, and Blender; produced low- and high-fidelity prototypes, physical device models, and renderings.
- Built a physical device model and conducted prototype testing on hand feel, portability, and interaction naturalness; iterated designs based on user feedback.
- Designed non-face-to-face interaction for introverted users and created an original five-dimensional interest taxonomy to enable stable social matching.

INTERNSHIP EXPERIENCE

National Institute of Biological Sciences, Beijing & Tsinghua Institute of Multidisciplinary Biomedical Research

Joint Research Intern *Jun 2026 – Aug 2026*

- Compared FACS-based facial behavior analysis tools (OpenFace 2.0/3.0, Py-Feat, LibreFace), evaluating AU, landmark, gaze, and emotion recognition pipelines for technical selection.
- Evaluated facial/emotion datasets (DAiSEE, RAER, RAVDESS, Aff-Wild2, DFEW, FERV39K) based on Confusion labels, AU annotations, video continuity, and context.
- Built a Py-Feat video-to-frame pipeline to extract AU, landmark, and related facial features from high-quality, high-intensity confusion samples.
- Explored various theoretical frameworks for facial recognition of emotions (such as FACS), computed an AU contribution matrix as prior weights, and trained a prototype Confusion recognition model using a self-curated dataset; supported ongoing iteration.

TECHNICAL SKILLS

Programming: Python, Java, C#, C/C++, JavaScript, HTML

AI/ML: PyTorch, RoBERTa, personality modeling, personalized modeling

AI Agents & LLM: Prompt engineering, RAG, LLM workflow design, LangChain, LangGraph

HCI & Affective Computing: Affective computing, multimodal interaction, emotion recognition, user research, usability testing

Hardware & Prototyping: Arduino, ESP32, STM32, sensor programming, 3D printing, product prototyping

Design & Tools: Figma, Blender, Procreate, Unity, SQL, Py-Feat, OpenFace, Cursor, Codex

ADDITIONAL

Activities: Researcher, Hybrid Interactive Intelligence Lab (HII Lab), Duke Kunshan University (Oct 2024 – Present); Member, Computer Science Club (Sep 2023 – Present)

Interests: Oboe (Grade 9), Piano